

Steps to find best model for the health insurance data set:

1. Look at the trend of the data for each of the variables by plotting each of the 6 predictor variables with the response variable "charges". For the quantitative variables scatterplots were used and for the categorical variables boxplots were used.

RESULTS

- a. The variables that would benefit most from interaction models are smoker, bmi, and age (also shown as the three variables in the full model with all 6 variables that have the lowest p values)
- b. The variables that would benefit from a quadratic term are age and body mass index
- c. The graphs show health insurance increases with smoking and higher bmi.

2. Make and compare first order models and interaction models. The models were compared using the anova function (nested tests) in R where if the p value is higher than the alpha level then the interaction term(s) is not useful in the models.

RESULTS

The best model from this step: charges = smoker + bmi + age + age * bmi + bmi * smoker + age * smoker + age * bmi * smoker (pval<2.2e-16, rsq=0.8368, adjrsq=0.836)

3. Create a full interaction model and reduce using stepwise regression.

RESULTS

The best model from this step: charges = sex + bmi + children + smoker + region + sex*children + bmi*smoker + bmi*region (pval<2.2e-16, rsq=0.7524, adjrsq=0.7501, AIC=23316.6)

4. Perform backwards elimination, forward selection, and stepwise regression tests in R.

RESULTS

All steps result in this first order model: charges = age + bmi + children + smoker + region (pval<2.2e-16, rsq=0.7509, adjrsq=0.7496, AIC=23314.58)

5. Fit a model with all the predictors plus the two quadratic terms (and some others not shown in the code because none of the reduced models with less predictors were any more useful)

RESULTS

The best model from this step: charges = age + bmi + children + smoker + region + (age^2) + (bmi^2) (pval<2.2e-16, rsq=0.7547, adjrsq=0.7528)

6. Perform best subset selection using r-sq, adjusted r-sq, mallows cp, bic, aic. All showed in scree plots that the most useful model has 3 variables. The top three predictors are smoker, age, and bmi respectively, consistent with initial predictions.

RESULTS

The best model from this step: charges = smoker + age + bmi (pval<2.2e-16, rsq=0.7475, adjrsq=0.7469)

7. Use cross validation on the models selected from each step and record their mean square value.

RESULTS

The best model and the one with the lowest ms value by a VERY significant amount is:
charges = smoker + bmi+ age + age * bmi + bmi *smoker + age* smoker + age * bmi * smoker
(pval<2.2e-16, rsq=0.8368, adjrsq=0.836,AIC=22748.55)

FINAL MODEL

Charges = -1.483e+03 - 1.242e+04(smokeryes) - 1.943e+01(bmi) + 2.458e+02(age) + 6.835e-01(bmi*age) + 1.184e+03(smokeryes*bmi) - 2.068e+02(smokeryes*age) + 6.624e+00(smokeryes*age*bmi) all interpretations of coefficients are at the bottom of the code document